PROJECT: Analysis of Breakage Angle of Chocolate Cakes

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In this project, I conducted a thorough analysis of the cake experiment data using Bayesian techniques to understand how the cake's texture and flexibility change as key baking factors, such as temperature and batch/replicate vary.

The sections of the report are as follows:

1) Data Description
2) Objective of this Analysis
3) Exploratory Data Analysis (EDA)-To Understand the data and select asuitable Bayesian model for this analysis
4) Model Framework(Bayesian Hierarchical Linear Model (BHLM))
5) Model Output Analysis and conclusion
6) Quick Analysis Summary for Executives
7) Bibliography

DATA DESCRIPTION

An experiment was conducted to collect data and analyze how different conditions (recipe, replicate, temperature, etc.) affect the cake's flexibility/texture.

The dataset contains the breakage angle of chocolate cakes made with three different recipes (A, B, C) and baked at six different temperatures. Each recipe has 15 replicates, where each replicate represents a batch. The replicate numbering reflects the temporal order in which the batches were produced.

Data Format

Variable Name	Type	Values	Description
recipe	Categorical	A, B, C	A, B, C are different cake recipies used in experiment
replicate	Categorical	1 to 15	Each recipe was baked 15 times (15 independent Batch). **Replicate = batch.**
temperature	Numeric	175, 185, 195, 205, 215, 225	Oven temperature (°F) at which the cake was baked. Each replicate is tested at 6 different temperatures.
angle	Numeric	Numeric values	The angle at which the cake collapses. Measures how much the cake sinks or falls apart after baking.

There are total 270 observations on the following 4 variables.

Note:

1) The replicate factor is nested within the recipe factor, and temperature is nested within replicate.

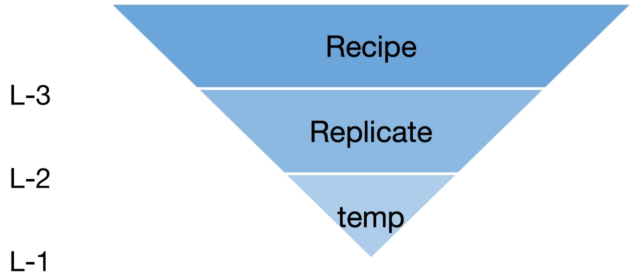
2) Even though the cakes were baked at only six different temperatures, for the purposes of my further analysis in this project, I will treat temperature as a continuous variable.

OBJECTIVE:

The objective of of this analysis are:

1. To identify how does temperature affect cake flexibility? the relationship between baking temprature of and cakes breakage angle.
2. Do different recipes respond differently to temperature?
3. Do replicate batches introduce variability? are all 15 replicates of a recipe behaved identically?

Exploratory Data Analysis (EDA) to assess the statistical properties of Cake experiment data



The data is at 3 level

Level-3: Recipe(A, B, C)

Level-2: Replicate nested in recipe

Level-1: Temparture nested in replicate

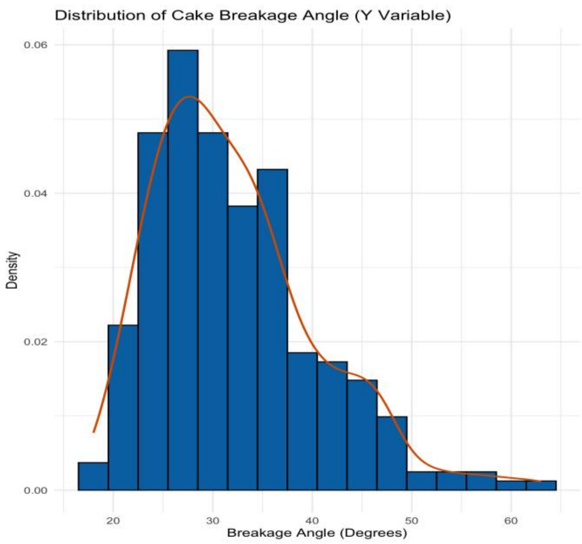


CHART-1

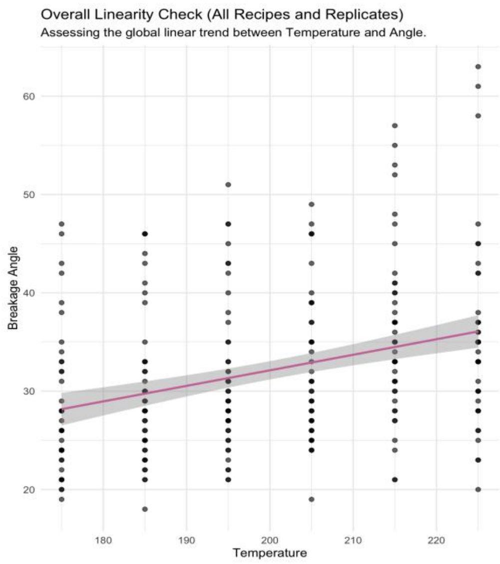


CHART-2

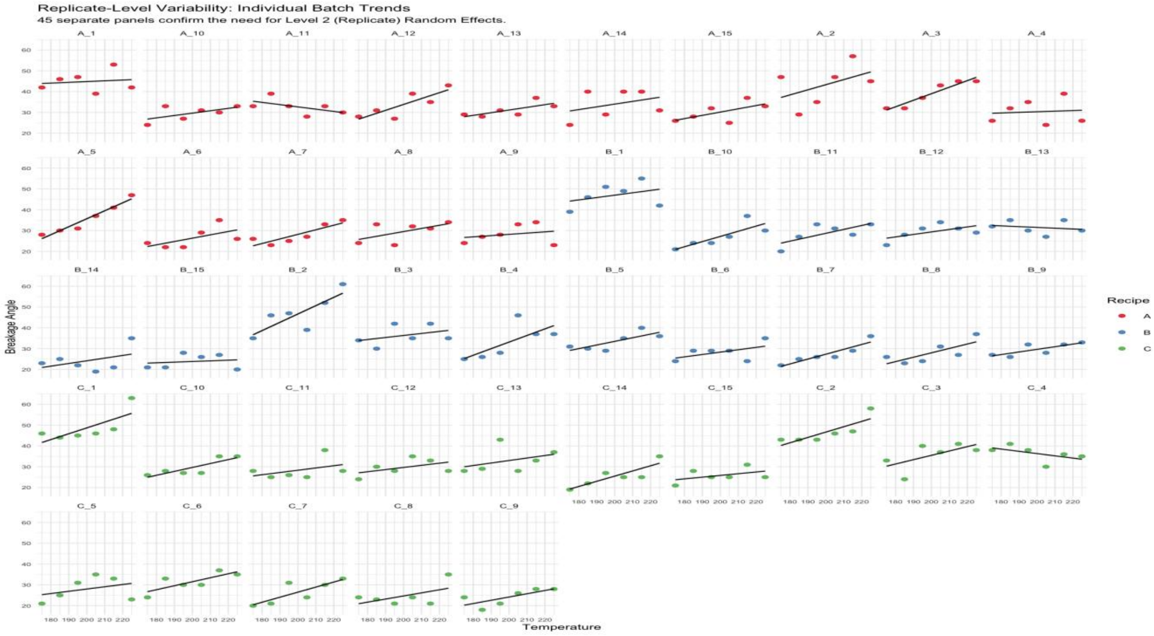


CHART-3

Findings from Charts

Chart 1-Distribution of Angle (Y): Since the frequency distribution of the cake breakage angle is close to a normal distribution, no transformation of the response variable (Y) is required for the **Bayesian Hierarchical Linear Model (BHLM)**.

Chart 2- Overall Angle (Y) vs. Temperature (X) : The experiment uses only six distinct temperatures, which results in a discrete set of points along the X-axis. However, the overall pattern shows a linear relationship between angle and temperature with a positive slope. This indicates that the cake breakage angle increases as the baking temperature increases.

Chart 3-Angle (Y) vs. Temperature (X) by Replicate and Recipe: The relationship between the angle and temperature for each batch (Replicate) within each Recipe is linear. However, the slope and the starting point (intercept) of this relationship are varying across the different replicates.

Exploratory Data Analysis (EDA) Conclusion and Model Selection:

The cake experiment data is Hierarchical and the response variable " Y" is normally distributied. The relationship of **Temperature and angle is** linear at all levels, and both the slope and the intercept vary randomly across the Recipe and Replicate levels. Based on this information, the Bayesian model I choose for the cake data is a " **Bayesian Hierarchical Linear Model (BHLM)" with varying slopes and varying intercepts.** This model will correctly account for all sources of variation.

Model Framework(Bayesian Hierarchical Linear Model (BHLM))

The response variable is the **Angle** (Y)

The primary predictor is **Temperature** (X)

$$\text{Angle} = \text{intercept} + \beta \times \text{temperature} + \text{error}$$

Let:

i: Index for Recipe (Level 3)

k: Index for Replicate (Level 2, nested within i)

j: Index for Temperature (Level 1)

The observed angle Y_{ijk} is assumed to be normally distributed

$$Y_{ijk} \sim Normal(\mu_{ijk}, \sigma^2)$$

The mean angle μ_{ijk} is defined by the intercept and slope estimated for that specific Replicate k within Recipe i:

$$\mu_{ijk} = \beta_{0,ik} + \beta_{1,ik} \times Temperature_j$$

Where:

$\beta_{0,ik}$: The Varying Intercept

$\beta_{1,ik}$: The Varying Slope

PRIORS

These equations define the full hierarchical structure, showing how the intercepts and slopes are linked across the three levels.

Level A: Global Priors

Non-informative priors are assigned to the global mean intercept and global mean slope.

Global Mean Intercept $\gamma_{00} \sim Normal(0, \sigma_{int}^2) = Normal(0, 100^2)$

Global Mean Slope $\gamma_{10} \sim Normal(0, \sigma_{slope}^2) = Normal(0, 100^2)$

Level B: Recipe Priors (β_i)

The recipe-level parameters (β_{0i} and β_{1i}) are drawn from distributions centered on the global parameters so that the expected slope and intercept for each recipe i shrink toward the Global Mean. This will estimate the true difference between recipes.

Recipe i Mean Intercept $\beta_{0i} \sim Normal(\gamma_{00}, \sigma_{Recipe0}^2)$

Recipe i Mean Slope $\beta_{1i} \sim Normal(\gamma_{10}, \sigma_{Recipe1}^2)$

$\sigma_{Recipe0}^{2}$ is the variance between the recipes' mean intercepts

$\sigma_{Recipe1}^{2}$ is the variance between the recipes' mean slopes

Level C: Replicate Priors (β_{ik})

The actual slope and intercept for each Replicate k within Recipe i shrink toward the Recipe i mean. This is where the Replicate effect is captured.

$$\begin{pmatrix} \beta_{0,ik} \\ \beta_{1,ik} \end{pmatrix} \sim MVN\left(\begin{pmatrix} \beta_{0i} \\ \beta_{1i} \end{pmatrix}, \Sigma_{Replicate}\right)$$

$$\Sigma_{Replicate} = \begin{pmatrix} \sigma_{Rep0}^{2} & \rho\,\sigma_{Rep0}\,\sigma_{Rep1} \\ \rho\,\sigma_{Rep0}\,\sigma_{Rep1} & \sigma_{Rep1}^{2} \end{pmatrix}$$

Priors Used in the Gibbs Sampler

The weakly informative priors used for the model parameters are:

1. **Global Mean (γ):** Normal prior N(0, 100I), where I is the 2 × 2 identity matrix

I have assigned a very large prior value for variance so that it's non-informative and to let the data decide

$$\gamma \sim N(0, 10^{4}I)$$

2. **Residual Variance (σ_y^2):** Inverse-Gamma prior Inverse-Gamma(a0, b0)

I choose non-informative Inverse-Gamma distribution to ensure the variance estimate is positive and driven almost entirely by the data.

$$\sigma_y^{2} \sim \text{Inverse-Gamma}(0.001, 0.001)$$

3. **Recipe Covariance (Σ_{recipe}):** Inverse-Wishart prior

Inverse-Wishart is the best choice for this prior as this is conjugate prior and minimally informative

$$\Sigma_{recipe} \sim IW(K+1, I)$$

where I = Identity matrix and K = 2 since we are estimating 2 parameters, intercept and slope

4. **Replicate Covariance (Σ rep):** Inverse-Wishart prior

Inverse-Wishart is the best choice for this prior as this is conjugate and minimally informative

$$\Sigma_{rep} \sim IW(K+1, I)$$

where I = Identity matrix and K = 2 since we are estimating 2 parameters, intercept and slope

MODEL OUTPUT ANALYSIS

NOTE: The response variable Temperature ("Y") is centered to its mean value (200 ^{0}F) in the data before model application.

Global level analysis

Parameter	Mean Value	95% Credible Interval
Global Mean Intercept	32.169	[29.892 , 34.408]
Global Mean Slope	0.1514	[-0.62787 , 0.98358]
Residual Variance (σ^2)	20.107	[16.482 , 24.697]

Findings and Conclusion:

1. The global mean intercept represents the general intercept value across all recipes (A, B, and C), ignoring the specific differences between them.
2. The global mean intercept value is 32.17, and the confidence interval is narrow, which tells that the model is confident in this global average.
3. The global mean intercept is 32.17, which means the overall expected angle for any generic replicate, baked at the average temperature 200 ^{0}F, is approximately 32.17 units.
4. On average, a one-unit increase in temperature will increase the angle by 0.15 units across all recipes.
5. The 95% confidence interval contains zero, which means we cannot confidently say if the temperature has a positive, negative, or no effect on the angle at global level.
6. The residual standard deviation is σ = 4.48$. This indicates that, on average, individual angle measurements are expected to deviate by approximately 4.48 units from their respective predicted values, even when controlling for all model parameters.

Recipe level analysis

Parameter	Mean Value	95% Credible Interval
Recipe A Slope	0.14966	[0.03864 , 0.26051]
Recipe B Slope	0.16604	[0.05164 , 0.28014]
Recipe C Slope	0.1566	[0.04429 , 0.26976]
Recipe A Intercept (Angle @ Mean Temp)	32.3	[30.006 , 34.595]
Recipe B Intercept (Angle @ Mean Temp)	32.093	[29.799 , 34.268]
Recipe C Intercept (Angle @ Mean Temp)	32.088	[29.798 , 34.312]
Recipe-Level Variance in Intercept	0.872	[0.117 , 3.861]
Recipe-Level Variance in Slope	0.46732	[0.0905 , 1.88019]

Posterior density plot of Global mean and Recipe(A, B, C)

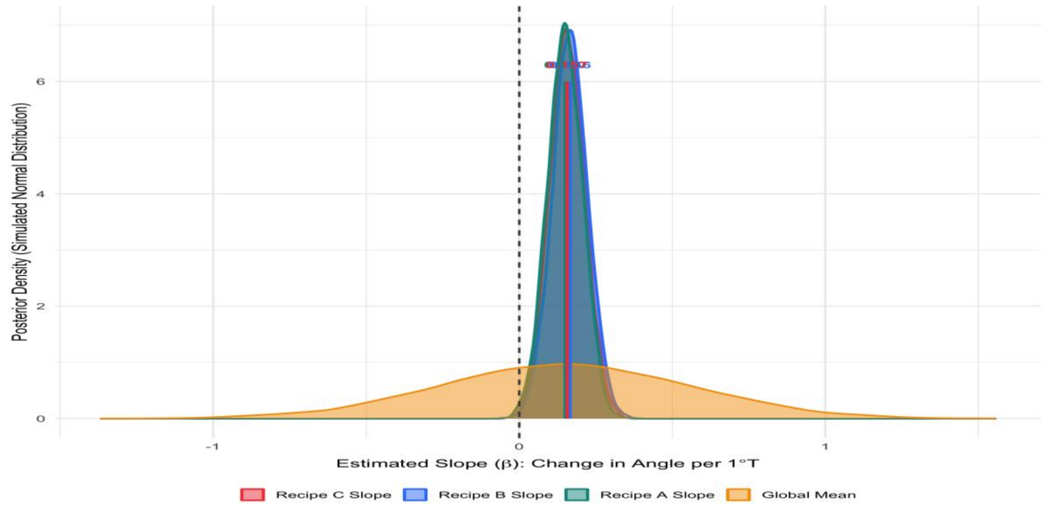


Chart-4

Findings and Conclusion:

1. The mean slopes of all three recipes are positive, and their credible intervals are narrow and exclude 0. This indicates that the breakage angle for all three recipes increases as temperature increases.
2. However, the credible intervals of the slopes overlap heavily, so we cannot clearly determine which recipe is more or less sensitive to temperature changes.
3. The intercepts of all three recipes are almost the same, meaning the expected breakage angle at the mean temperature (200 ˚ F) is nearly identical across all recipes.
4. Overall, we cannot conclude that the different recipes have meaningfully different impacts on breakage angle as temperature changes.
5. The recipe-level variances for both intercepts and slopes are small. While the recipes are not exactly the same, the differences between them are not significant enough to be considered meaningful.

Replicate Level analysis

Parameter	Mean	95% Credible Interval
Replicate-Level Variance (Intercept)	40.153	[25.372 , 62.024]
Replicate-Level Variance (Slope)	0.03996	[0.0257 , 0.0626]

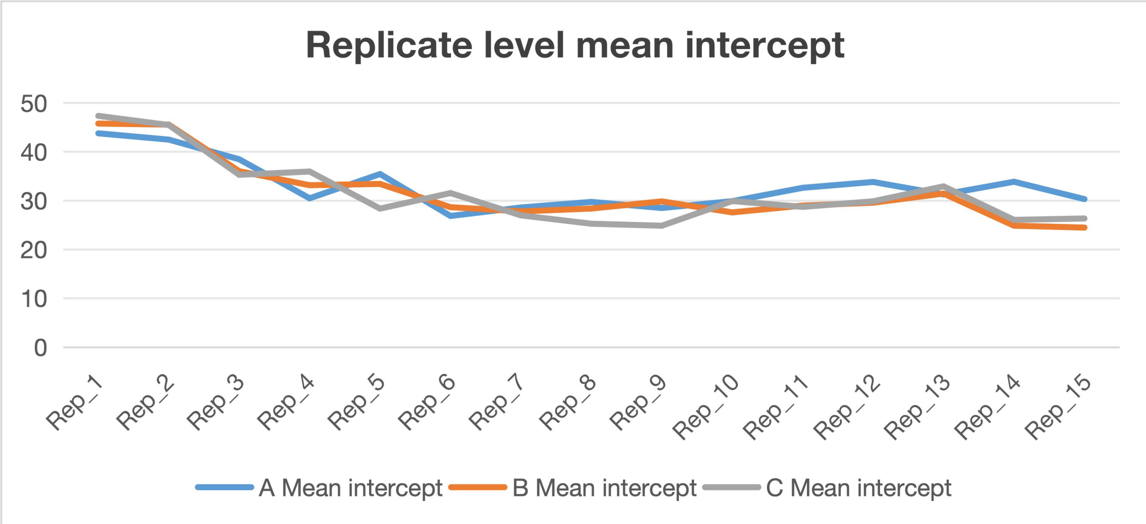


Chart-5

Findings and Conclusion:

1. The replicate/batch-level intercept variance within each recipe is very high (40.143), indicating substantial replicate-to-replicate variability. This means that even with the same recipe and temperature, different batches can bake caked with different breakage angle.
2. The intercept values are highest for the initial replicates (Replicates 1, 2, and 3) and then decrease, as shown in **Chart-5**. This suggests that, for a given recipe, cakes baked in the early batches tend to be more flexible (higher breakage angle).
3. The replicate-level slope variance is low, indicating that the slopes across replicates are very similar. In other words, for a 1-unit increase in temperature, the change in breakage angle is nearly the same across all replicates.

Overall Conclusion:

1. Temperature positively affects cake flexibility, and this effect is consistent across all recipes.
2. The recipe-level variances for both intercepts and slopes are small. Although the recipes are not exactly the same, the differences between them are not statistically meaningful.
3. Replicate-level differences are large for intercepts but small for slopes. This means that different batches produce very different baseline breakage angles, but they respond similarly to changes in temperature.

Quick Analysis Summary for Executives
1. Recipes A, B, and C behave similarly, showing nearly identical sensitivity to temperature. 2. Bakers are not constrained to a specific recipe for cake’ s structural designs; flexibility of cake sponge can be adjusted by modifying baking temperature and batch of baking. 3. For designs requiring curved or flexible cake layers, baking at higher temperatures and using early batches will give the best cake sponge for cuve shaping. 4. For tall, layered cakes that require firmer, less flexible sponges, bakers can achieve better structure by baking at comparatively lower temperatures and using batter from later batches. This produces sturdier cake layers that maintain their shape in tall constructions. 5. The variation across replicates could be related to batch order, but this is only true if all external factors remained constant. Since **correlation does not imply causation**, the observed differences cannot be solely attributed to replicate ordering. 6. Other factors may also contribute to replicate-level variability, such as measurement error, changes in oven conditions across batches, or differences in batter quantity etc. To identify the true causes, further experiments can be done to analyse and confirm a reson.

BIBLIOGRAPHY

The cake experiment description link: https://rdrr.io/cran/lme4/man/cake.html
Experiment Data link: https://vincentarelbundock.github.io/Rdatasets/csv/lme4/cake.csv
Source: Original data were presented in Cook (1938), and reported in Cochran and Cox (1957, p. 300). Also cited in Lee, Nelder and Pawitan (2006).